Reducible cube dynamics on regular contractible sets in four dimensions

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Abstract

We consider the topology-preserving cube dynamics in the higher-dimensional extension of the Ising-disk model. The states are all finite regular contractible unions of lattice cubes containing a fixed cube. A removal must admit a strong deformation retraction that keeps the removed cube inside itself, and additions are the inverse moves. In four dimensions, every state is a piecewise-linear manifold with boundary, and every permitted move preserves the homeomorphism type of that boundary. A contractible four-manifold with nontrivial boundary fundamental group, embedded by Lickorish in the four-sphere, can be realized exactly as such a lattice-cube union by a hypersurface cubulation theorem. It therefore cannot communicate with the state consisting of the fixed cube alone. This gives a negative answer to the proposed universal higher-dimensional extension. The obstruction holds for every choice of positive transition rates; the separate three-dimensional question is not settled here.

1 The model and the obstruction

Baryshnikov and Onaran proved irreducibility and small-parameter ergodicity for their planar Ising-disk dynamics and asked for a higher-dimensional extension [1, Theorem 6.5 and Conjecture 6.9]. We use the precise all-state formulation of AIM Problem 500[1] and retain its local retraction requirement.

For $d \geq 2$ and $z \in \mathbb{Z}^d$, write

$$Q_z = z + [-1/2, 1/2]^d.$$

A finite collection C of these cubes is a *clump* if its union $|C|$ is contractible and regular. Regularity means that every $x \in |C|$ is either an interior point, in the ambient Euclidean space, or

$$(B_\varepsilon(x) \cap |C|) \setminus \{x\} \quad \text{is contractible for every sufficiently small } \varepsilon > 0. \tag{1}$$

The state space $\mathcal{C}_d$ consists of all clumps containing Q_0. If $Q \in C \setminus \{Q_0\}$, its removal is permitted when $D = C \setminus \{Q\}$ is a clump and there is a strong deformation retraction $H : |C| \times [0,1] \longrightarrow |C|$ onto $|D|$ such that

$$H(Q \times [0,1]) \subseteq Q. \tag{2}$$

Thus H fixes $|D|$ at every time. Addition is the inverse of a permitted removal. Every permitted addition has rate $p_\uparrow \in (0,1]$, and every permitted removal has rate $p_\downarrow \in (0,1]$.

[1] The frozen problem statement is AIM Problem 500 at revision `1b7c7fd`.

Theorem 1.1. *For the dynamics above in dimension $d = 4$, a permitted move preserves the homeomorphism type of the boundary of the cube union. There exists $C_* \in \mathcal{C}_4$ such that*

$$\pi_1(\partial|C_*|) \neq 1.$$

Consequently, C_* and $\{Q_0\}$ belong to different components of the move graph. For every $p_\uparrow, p_\downarrow \in (0, 1]$, the continuous-time chain is not irreducible on all of $\mathcal{C}_4$.

Here ∂ denotes ambient boundary and π_1 the fundamental group. All boundaries in the theorem are connected compact three-manifolds, so their fundamental groups have an unambiguous isomorphism type. The move graph has the states as vertices and an edge for each permitted addition–removal pair. Its components use finite paths, with no restriction to a fixed bounding box.

The theorem refutes the assertion quantified over every $d \geq 2$. It does not contradict the planar result, and it leaves the $d = 3$ case open. The construction is an existence argument: no explicit finite list of cube coordinates or numerical certificate is needed or claimed.

2 Regular cube unions in four dimensions

We use integral homology throughout. A nonempty space is called *acyclic* when all its reduced integral homology groups vanish. The piecewise-linear (PL) structure on a cube union comes from a compatible triangulation of its cubes and faces.

2.1 Finite links and embedded low-dimensional manifolds

The local links below are finite polyhedra with compatible triangulations. For subcomplexes A, B of a finite complex $X = A \cup B$, the usual reduced Mayer–Vietoris sequence applies, including when A and B are closed. For clarity, this follows directly from its open-cover form [3, pp. 149–151]. In the barycentric subdivision, let U_A be the set where the total coordinate weight on vertices representing faces in A is positive, and define U_B similarly. No simplex contains both an A-only vertex and a B-only vertex: its vertices represent a nested chain of faces, and membership in a subcomplex is downward closed. The sets U_A, U_B form an open cover and retract onto A, B by deleting the other weights and renormalizing. In $U_A \cap U_B$ the weight on common vertices is positive; the same operation retracts the intersection onto $A \cap B$. This gives the claimed sequence. Applied to the two cones of a suspension, it also gives

$$\widetilde{H}_{j+1}(\Sigma L; \mathbb{Z}) \cong \widetilde{H}_j(L; \mathbb{Z}). \tag{3}$$

We use this degree shift for iterated suspensions.

Lemma 2.1. *Let $n = 2$ or 3. If a compact connected acyclic PL n-manifold K with boundary is PL embedded in S^n, then K is a PL n-ball.*

Proof. The codimension-zero embedding makes K orientable. Its boundary is nonempty, since a closed connected orientable n-manifold has nonzero top homology. The universal coefficient theorem [3, Theorem 3.2, p. 195] and acyclicity give $H^0(K; \mathbb{Z}) = \mathbb{Z}$ and $H^j(K; \mathbb{Z}) = 0$ for $j > 0$. Poincaré–Lefschetz duality [3, Theorem 3.43, p. 254] therefore gives

$$H_i(K, \partial K; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & i = n, \\ 0, & i \neq n. \end{cases}$$

The exact sequence of the pair shows that ∂K is connected and has the integral homology of S^{n-1}. If $n = 2$, this boundary is a polygonal circle. If $n = 3$, it is a closed connected orientable PL surface with $H_1(\partial K; \mathbb{Z}) = 0$, hence a PL two-sphere.

One can verify the last surface assertion without any three-manifold classification. In a triangulated orientable closed surface, a spanning tree of the one-skeleton has a disk neighborhood. Contracting the tree replaces that neighborhood by a disk with the same boundary and leaves the PL surface type unchanged. The dual graph of the remaining edges is connected: removing the interior of the tree neighborhood leaves a connected surface. Merging faces across a dual spanning tree gives a cell presentation with one vertex, one face and e edges. Before its remaining side identifications, the face is a disk, formed by adding triangles along one edge at a time. Each remaining edge occurs once in each direction in the oriented face boundary. Both cellular boundary maps vanish, so $H_1 = \mathbb{Z}^e$ [3, Theorem 2.35, pp. 138–139]. Thus $H_1 = 0$ forces $e = 0$, giving a disk with its boundary collapsed to a point, which is a PL two-sphere.

The PL Schoenflies theorem in dimensions two and three now shows that each closure of a component of $S^n \setminus \partial K$ is a PL n-ball [6, pp. 47–49]; see also the explicit PL formulation in [4, pp. 1–2]. The interior of K is one whole complementary component: it is connected and is both open and closed relative to that component, since its only frontier is ∂K. Hence K is its ball closure. □

Proposition 2.2. *Every finite regular union of four-dimensional lattice cubes is a compact PL four-manifold with boundary. If the union is also contractible, its boundary is a connected integral homology three-sphere.*

Proof. Let x lie in the relative interior of a k-dimensional lattice face, and put $r = 4 - k$. For $k < 4$, a neighborhood of x in the union is a product of $\mathbb{R}^k$ with a union of coordinate orthants in $\mathbb{R}^r$. Its normal link L is a nonempty pure $(r-1)$-dimensional subcomplex of the boundary of the r-dimensional cross-polytope. If all orthants occur, x is interior and $L = S^{r-1}$. Otherwise, x is a boundary point, and its full local link is

$$S^{k-1} * L = \Sigma^k L,$$

with $k = 0$ interpreted as L itself. Small punctured ball intersections retract onto this link. Regularity and (3) imply that L is acyclic.

We check the normal links in increasing order of r. For $r = 1$, a proper nonempty link is a point. For $r = 2$, an acyclic subcomplex of a polygonal circle is an interval. For $r = 3$, each vertex link in L is the $r = 2$ normal link at a nearby lattice-face point, hence an interval or a circle. Thus L is a compact PL surface with boundary embedded in S^2. Lemma 2.1 makes it a disk. For $r = 4$, the vertex links are the $r = 3$ links just considered, hence disks or two-spheres. Consequently, L is a compact PL three-manifold with boundary in S^3, and Lemma 2.1 makes it a three-ball. At nearby points, a full normal link is a sphere and a proper one is treated by regularity; these are exactly the two possibilities in the induction.

Every boundary normal link is therefore a PL ball, and every interior normal link a sphere. The local product-cone description yields the PL manifold charts. Compatible triangulation also covers points inside the subdivided cells. Points in cube interiors are already ordinary interior points. Finally, for a contractible union, duality and the pair exact sequence, as in Lemma 2.1 with $n = 4$, give the stated boundary homology. This last step asserts nothing about the boundary fundamental group. □

3 A move replaces a boundary ball by a boundary ball

First we show why a permitted cube attachment has no isolated lower-dimensional attachment faces. Both states must be regular for this argument.

Lemma 3.1. *Let D and $C = D \cup \{Q\}$ be finite regular cube collections in $\mathbb{R}^d$, where $Q \notin D$. Suppose there is a strong deformation retraction of $|C|$ onto $|D|$ satisfying (2). Then*

$$A = Q \cap |D|$$

is nonempty, contractible, and a union of full $(d-1)$-facets of Q.

Proof. Restrict the homotopy to Q. Its final image lies in $Q \cap |D|$, and it fixes this intersection at all times. This is a strong deformation retraction of Q onto A, proving nonemptiness and contractibility. Because lattice cubes meet in full faces, A is a union of faces of Q.

Suppose F is a maximal face of this union with dimension $k < d-1$, and choose x in its relative interior. In a sufficiently small ball about x, the intersection $Q \cap |D|$ is exactly F. The point is not interior to $|D|$, which misses the interior of Q. It is not interior to $|C|$ either: filling a neighborhood would require the cube across each facet of Q containing F to belong to D, contrary to maximality of F.

Intersect the local cones at x with a small sphere. Their links satisfy

$$L_C = L_Q \cup L_D, \qquad L_Q \cap L_D = S^{k-1}.$$

The links L_Q and L_D are contractible; regularity would also make L_C contractible. If $k \geq 1$, the finite-subcomplex Mayer–Vietoris sequence established above gives

$$\widetilde{H}_k(L_C; \mathbb{Z}) \cong \widetilde{H}_{k-1}(S^{k-1}; \mathbb{Z}) = \mathbb{Z},$$

a contradiction. If $k = 0$, the intersection is empty and the two nonempty links make L_C disconnected. Thus every maximal attachment face is a facet. □

Lemma 3.2. *A nonempty union of facets of a four-cube is a compact PL three-manifold with boundary, possibly disconnected.*

Proof. At a point on $r \geq 2$ incident facets of the cube, the normal link in its boundary is the boundary of an $(r-1)$-simplex. The normal link in the selected facet union is the corresponding union of simplex facets. A full selection is a sphere; any proper nonempty selection is a ball. To see the latter assertion by induction, choose a vertex opposite an unselected facet. Every selected facet contains this vertex, so the union is its cone on a union of facets of the opposite simplex. The base is either that simplex's whole boundary or a proper nonempty union; by induction it is a sphere or a ball, and its cone is a ball. The initial case is one selected endpoint of an interval. At $r = 1$, the point is in the interior of a selected cube facet and has an ordinary $\mathbb{R}^3$ chart. These links prove the assertion at all face dimensions. □

Proposition 3.3. *If two four-dimensional clumps differ by a permitted move, their boundaries are homeomorphic.*

Proof. Write the two states as D and $C = D \cup \{Q\}$. Proposition 2.2 makes both unions compact PL four-manifolds. By Lemma 3.1, $A = Q \cap |D|$ is a contractible union of facets. Lemma 3.2 and Lemma 2.1, applied in $\partial Q \cong S^3$, show that A is a three-ball. It is proper in ∂Q, since a three-sphere is not contractible. Its boundary is a PL two-sphere, and PL Schoenflies shows that

$$B = \overline{\partial Q \setminus A}$$

is the complementary three-ball, with $A \cap B = \partial A = \partial B$.

The set A lies in $\partial |D|$, since $|D|$ contains no interior point of Q. Let A° be the intrinsic interior of the three-ball A. Invariance of domain in local three-dimensional charts makes A°

open in $\partial|D|$ [3, Theorem 2B.3, p. 172]. The PL two-sphere ∂A is locally flat there: its local links are polygonal circles in the two-sphere links of the three-manifold. The boundary after the attachment is therefore

$$\partial|C| = (\partial|D| \setminus A^\circ) \cup B. \tag{4}$$

Here is a pointwise check of the formula. At a point of A°, the boundary patches of $|D|$ and Q agree locally. Their manifold interiors occupy the opposite sides, so this point becomes interior to $|C|$. This uses the openness of the common patch and the manifold charts, not just the list of incident selected facets. At a point of ∂A, at least one incident facet of Q is unselected. The neighboring cube across that facet is absent from D, leaving an open complementary sector arbitrarily close to the point; it remains on the boundary of $|C|$. Outside A, the boundaries of the two pieces are unchanged. This proves (4).

There is a homeomorphism $f : A \longrightarrow B$ equal to the identity on their common boundary. Indeed, identify each with a standard three-ball and extend the resulting boundary homeomorphism radially by Alexander's trick. On $\partial|D|$, use f on A and the identity on $\partial|D| \setminus A^\circ$. The maps agree on ∂A, and (4) shows that the result is a continuous bijection onto $\partial|C|$. Compactness makes it a homeomorphism. The argument applies to a removal in reverse and is independent of the choice of pinned cube. □

4 Realizing a contractible manifold with nontrivial boundary

We now construct an admissible state with a boundary different from that of a cube. The external input is an embedded contractible four-manifold, followed by a theorem that cubulates its boundary. The passage from that boundary to an exact union of whole cubes is included below.

Lemma 4.1. *There exists a clump $C_* \in \mathcal{C}_4$ with $\pi_1(\partial|C_*|) \neq 1$.*

Proof. Lickorish's Theorem 1 [5, pp. 284–286] constructs a compact contractible four-manifold with smooth or PL embeddings in S^4. Its proof explicitly establishes that the manifold's boundary fundamental group is nontrivial (p. 286). Choose a smooth embedding and denote its image by W. Removing a point outside W identifies the complement of that point in S^4 with $\mathbb{R}^4$. The image $W \subset \mathbb{R}^4$ remains compact and contractible, and

$$\pi_1(\partial W) \neq 1.$$

Its boundary $M = \partial W$ is a smooth compact closed three-dimensional hypersurface. The outward normal trivializes its normal bundle, and a small closed normal interval supplies a tubular neighborhood.

Apply Boege–Hinojosa–Verjovsky's hypersurface cubulation theorem [2, Theorem 2.8 in the author preprint, version 2] with $n = 2$. For a sufficiently fine subdivision of the canonical cubulation of $\mathbb{R}^{n+2}$, that theorem makes the union of cubes meeting a smooth compact closed M^{n+1} a bicollar, and gives an ambient isotopy carrying M onto either boundary component. The component is a union of whole $(n+1)$-faces of the fine cubulation. Let φ be the final ambient homeomorphism and set $W' = \varphi(W)$. Then W' is compact and contractible, its boundary is a finite union of three-faces of a grid of mesh $1/m$ for an integer m, and its boundary fundamental group is nontrivial. Because this is an ambient image of the whole W, it is also a locally flat codimension-zero topological manifold with boundary.

Every open four-cell of the fine grid is connected and disjoint from $\partial W'$, so it lies entirely inside or entirely outside W'. Take the closures of all cells whose interiors lie inside. There are finitely many by compactness, and their union is exactly W'. One inclusion follows from closedness of W'. For the other, W' is the closure of its interior, and interior points off the

grid skeleton are dense there; they belong to the selected cells. A finite union of their closures is closed and hence contains every point of W'.

We must check the Euclidean-ball condition (1), rather than only the existence of manifold neighborhoods. A finite polyhedron is locally conical at each point, so for a boundary point $x \in W'$, all sufficiently small punctured ball intersections have the same conical link. Fix one such radius ε. Choose an open manifold half-ball neighborhood U of x inside $B_\varepsilon(x) \cap W'$, and then $\delta > 0$ so small that $B_\delta(x) \cap W' \subset U$. The punctured inclusion

$$i : (B_\delta(x) \cap W') \setminus \{x\} \longrightarrow (B_\varepsilon(x) \cap W') \setminus \{x\}$$

is a homotopy equivalence by radial scaling in the local cone. It factors through $U \setminus \{x\}$, which is contractible because x is the boundary center of a manifold half-ball. Thus i is null-homotopic. Composing with a homotopy inverse shows that the identity on the smaller punctured intersection is null-homotopic, so both punctured intersections are contractible. The argument works for every sufficiently small ε. Interior points need no such check.

Finally, dilate by m to obtain unit cubes with centers in $(\mathbb{Z} + 1/2)^4$, translate by $(-1/2, -1/2, -1/2, -1/2)$, and then translate the center of any chosen cube to zero by an integer vector. These similarities preserve contractibility, the boundary group, and (1) by scaling Euclidean balls. The resulting finite collection C_* has precisely the required lattice centers and contains Q_0. It is the desired clump. □

Proof of Theorem 1.1. Proposition 3.3 proves boundary preservation under every permitted move. The boundary of the single cube Q_0 is S^3 and has trivial fundamental group; Lemma 4.1 gives C_* with nontrivial boundary fundamental group. A finite path cannot join them.

All permitted rates are positive, so the jump process uses exactly the same move graph. A size-N state has at most $(3^4 - 1)N$ candidate additions, since a new cube must meet an existing one, and at most $N - 1$ removals. The rates are at most one, so the exit rate is finite. Every finite jump history from $\{Q_0\}$ stays in its component. For the minimal jump process, explosion can kill a path but cannot introduce a transition into the component containing C_*. In particular, a nonexplosive chain of the kind requested by the problem cannot be irreducible on the complete state space. □

Remark 4.2. For $\beta = p_\uparrow / p_\downarrow > 0$, suppose

$$Z_4(\beta) = \sum_{C \in \mathcal{C}_4} \beta^{\#C} < \infty, \qquad \Pi_\beta(C) = Z_4(\beta)^{-1} \beta^{\#C}.$$

Here $\#C$ is the number of cubes in C. The all-state measure Π_β gives C_* positive mass. Starting from $\{Q_0\}$, the nonexplosive chain has probability zero of occupying C_* at every finite time. Its law therefore cannot converge to Π_β; its total variation distance is at least $\Pi_\beta(C_*)$. This conditional observation does not assert failure of normalization, nonexplosion, or stationarity of the all-state measure. It shows that none of those properties can repair the required all-state irreducibility and convergence by choosing a smaller positive β.

AI assistance. OpenAI Codex assisted with literature retrieval, development and checking of the argument, and preparation of the manuscript. The result is presented as a natural-language proof using the cited sources; independent mathematical review is invited.

References

[1] Y. Baryshnikov and E. Onaran, *Ising disks: topology preserving Glauber dynamics*, Journal of Applied and Computational Topology **10** (2026), article 16. doi:10.1007/s41468-026-00248-w.

[2] M. Boege, G. Hinojosa and A. Verjovsky, *Any smooth knot $S^n \hookrightarrow \mathbb{R}^{n+2}$ is isotopic to a cubic knot contained in the canonical scaffolding of $\mathbb{R}^{n+2}$*, Revista Matemática Complutense **24** (2011), 1–13. doi:10.1007/s13163-010-0037-4. Author preprint arXiv:0905.4053v2 (2009).

[3] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Author's text (teaching mirror).

[4] A. Hatcher, *Notes on Basic 3-Manifold Topology*, online notes. Author's text (teaching mirror).

[5] W. B. R. Lickorish, *Knotted contractible 4-manifolds in S^4*, Pacific Journal of Mathematics **208** (2003), 283–290. Publisher's text.

[6] C. P. Rourke and B. J. Sanderson, *Introduction to Piecewise-Linear Topology*, Springer-Verlag, 1972.